

Industrial Monitoring System Documentation

1. Overview

The Predictive Maintenance Dashboard is a comprehensive system designed to monitor heavy industrial machinery, predict breakdowns before they occur, and prescribe maintenance actions based on live calculated risks. The dashboard uses both a deterministic rule-based Risk Engine and a Machine Learning model.

2. Deterministic Risk Engine

The core of the system is the **Risk Engine**, which evaluates real-time telemetry and historical data to assign a quantifiable risk score to each machine.

Risk Score Calculation

The risk score (ranging from 0 to 100) is determined by four key factors:

1. Life Completed Risk (Maximum 30 points)

- Calculation: $(\text{Life Completed in Years} / 20.0) * 30$
- A machine is assumed to have an average maximum lifespan of 20 years. The closer it is to 20 years, the higher the risk score.

2. Breakdown Risk (Maximum 30 points)

- Calculation: $(\text{Progressive Breakdown Hours} / \text{Total Running Hours}) * 10 * 30$
- This measures how much of the machine's running time has been spent in a broken-down state.

3. Availability Risk (Maximum 20 points)

- Calculation: $100 - \text{Actual Progressive \% Availability}$
- Machines with lower availability % accrue higher risk penalties.

4. Utilization Risk (Maximum 20 points)

- Calculation: $100 - \text{Actual Progressive \% Utilization}$
- Under-utilized machines might indicate underlying operational issues.

The final Risk Score is the sum of these four factors, capped at a maximum of 100 points.

Risk Categories

Based on the final Risk Score, the machine is placed into a risk category:

- **HIGH RISK:** Risk Score ≥ 70
- **MEDIUM RISK:** Risk Score ≥ 40 and < 70

- **LOW RISK:** Risk Score < 40

3. Priority and Recommendations

Once a machine is categorized, the system assigns priorities and prescribes actionable steps:

- **HIGH RISK (Priority 1):**
 - **Status:** Machine is at imminent risk of failure.
 - **Recommendation:** Immediate Inspection.
 - **Next Action:** Schedule Immediate Inspection.
- **MEDIUM RISK (Priority 2):**
 - **Status:** Machine shows signs of wear or sub-optimal performance.
 - **Recommendation:** Service Within 7 Days.
 - **Next Action:** Plan Routine Service.
- **LOW RISK (Priority 3):**
 - **Status:** Machine is healthy.
 - **Recommendation:** Healthy.
 - **Next Action:** Continue Normal Operation.

4. Machine Learning Predictive Model

Alongside the rule-based Risk Engine, the system employs an AI model (Random Forest Classifier) to forecast absolute breakdown probabilities.

Model Features

The ML model trains on historical data using the following metrics:

- Progressive Work Hours
- Progressive Breakdown Hours
- Progressive Maintenance Hours
- Actual Progressive % Availability
- Actual Progressive % Utilization
- Total Running Hours
- Life Completed (in Years)

How it Works

1. The model identifies complex, non-linear patterns that lead to a machine entering the 'BREAKDOWN' state.
2. For each active machine, it outputs a **Breakdown Probability %** (e.g., 85%).
3. If a machine's probability exceeds 50%, it is heavily highlighted in the dashboard to alert operators before failure occurs.

5. Dashboard Features

- **Dynamic KPIs:** Real-time summary of Total Machines, Breakdowns, and counts across

High/Medium/Low risk bands.

- **Interactive Filtering:** Ability to drill down and isolate machines by **Area Description**, **Machine Name (Type)**, and **CIL Number**.
- **Advanced Visualizations:** Risk distribution charts, working status summaries, and histogram density plots for utilization and availability.
- **Intelligent Tables:** Sorted directory identifying the Top 20 most critical machines requiring immediate attention and the Top 20 healthiest machines.